



TRANSMISSION LINE PARAMETER

Conductors used in Transmission lines (overhead)

(a) Copper [expensive, so rarely used]

(b) ACSR [Aluminium conductor Steel reinforced]

- Mostly used bcoz Al is much cheaper & available in abundance.

- Aluminium has much lower conductivity as compared to copper which means we have to use larger cross sectional area for the same current carrying capacity of the conductor.

- Also its tensile strength is not very high. Therefore it needs to be reinforced.

(c) AAAR [All Aluminium alloy conductor]

- Higher tensile strength, expensive.

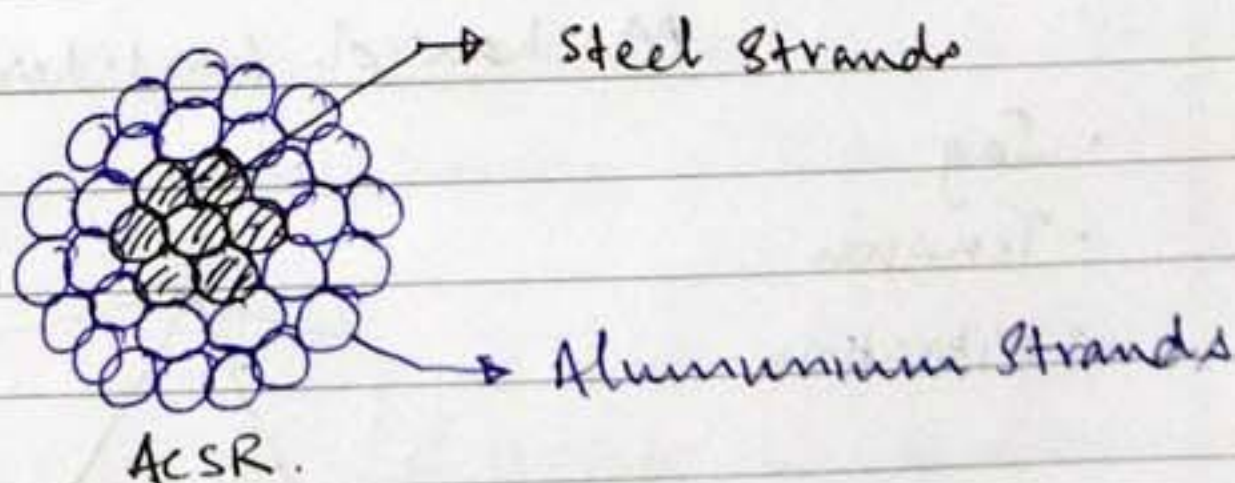
(d) AACR [Aluminium conductor Alloy reinforced]

(e) Expanded ACSR [used for extra high voltage line]

In this we provide some material ~~like~~ like filler material such as paper or fiber which increases the effective diameter of conductor for the same cross-sectional area.

~~also~~ (diameter $\propto$ extra stress reduce $\propto$ reduces corona losses)
reduces the inductance.

The conductors used are generally a stranded conductors:



Why Stranded Conductors?

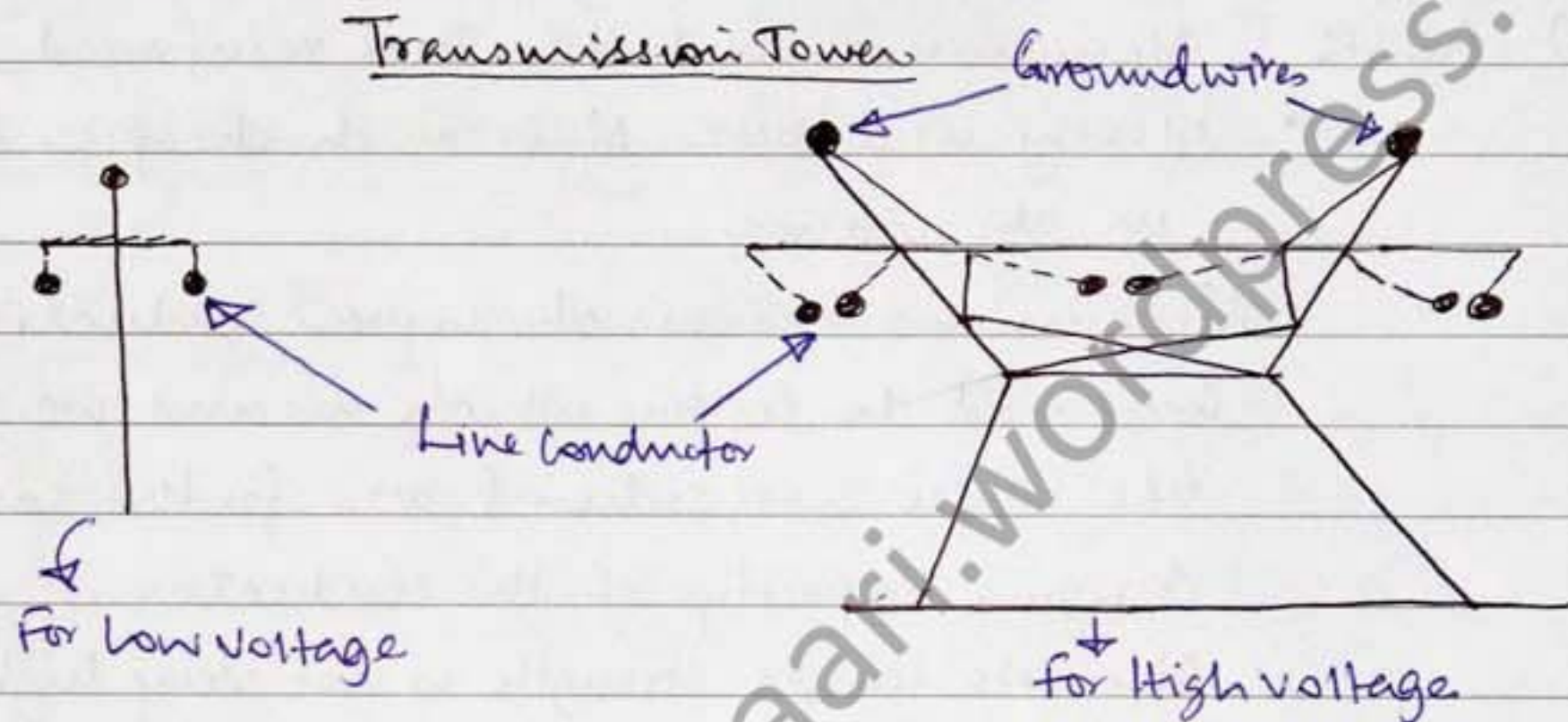
→ Manufacturing is easy (especially for larger sizes).

→ Provide better mechanical strength & Handling.

Therefore it is preferred over single solid conductor.

- we know Al has higher conductivity than steel. so, most of the current flows through the Aluminium.

Support Structure:



- For High voltage, we are using two or more conductor for each phase i.e. ~~But~~ Bundling of conductor is done. This is generally used above 220kV.

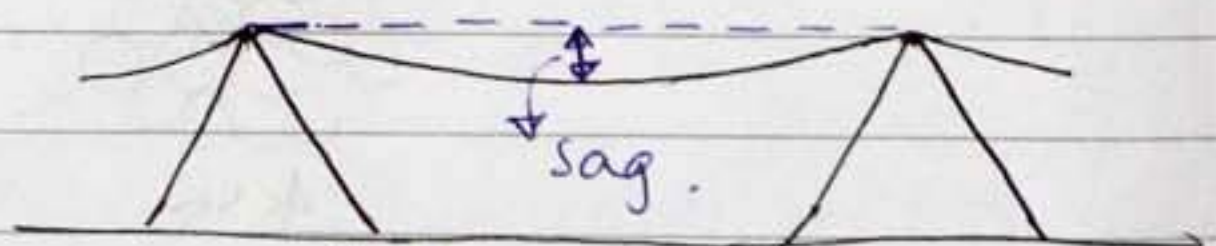
Reason for Bundling:

- increases the effective radius of conductor. Therefore, it reduces the electrical stress, corona losses and ~~inductance~~ inductance of conductor.

- Ground wire provide on top of tower to protect the live wire from Lightning Strikes.

Mechanical consideration

- Sag
- Tension
- Vibration



- Sag depends on the tension ~~at~~ that these conductors can take and therefore we cannot provide very large amount of tension to reduce sag.

- Another aspect of sag is that this governs what is the minimum ground clearance that is available (in order to protect the human or moving vehicle from voltage stress)
- "For 400kV line, we need atleast 12m. of ground clearance."
- Vibration occurs due to the wind flowing across the conductor which induces Fatigue in conductor & Tower and they become brittle and get damaged. Therefore, we need to damped this vibration. This can be done by adjusting the tension and by putting vibration damper on the conductors which absorbs the energy.
- For 400kV line, spacing b/w the conductor must be $\geq 10\text{m}$.

ELECTRICAL PARAMETER.

~~(A) Resistance~~

(A) Resistance

We are using overhead conductors which have some resistance to the current flowing through them. So, there is going to be power loss in these conductor. Therefore we must find out the resistance.

(b) Inductance

When current flows in the conductor, it sets up magnetic field b/w of which there is certain amount of inductance arises.

(c) Capacitance

Since these conductors are at high voltage, there is a voltage difference b/w the two phases or b/w the phases & ground. Therefore, there is capacitance involved b/w them.

RESISTANCE

$$\rightarrow \boxed{R_{d.c.} = \frac{\rho l}{A}}$$

ρ = resistivity ($\propto$ temperature)

l = Length of conductor

A = Cross-sectional area

$\rightarrow$ Resistivity of conductor metals varies linearly over normal operating temperatures according to

$$(\rho)_{T_2} = \rho_{T_1} \left(\frac{T_2 + T}{T_1 + T} \right)$$

T = temperature constant
& value is diff. for diff. material.

$\frac{1}{2}$ $\rightarrow$ The AC resistance is somewhat higher than the d.c. resistance. This is because at higher frequency, the current in the conductor is no longer uniform i.e. at higher frequency, less current is flowing through the center of the conductor & more current will flow through the outer part of the conductor. This is known as Skin Effect. This reduces the effective cross sectional area and therefore resistance increases.

$$\boxed{R_{a.c.} = \frac{P_{loss}}{I^2}}$$

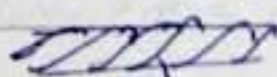
$\rightarrow$ So, Resistance of conductor depends on

(a) temperature

(b) frequency (skin effect)

$\rightarrow$ (c) Spiraling

Spiraling is done to provide the mechanical strength of conductor which $\uparrow$ the effective length & therefore $R \uparrow$.


 $\downarrow$ spiraling

INDUCTANCE

For calculating Inductance, we need to go through 4 steps:

1. Calculate Magnetic field Intensity H from Ampere's law.
2. Magnetic flux density $B = \mu H$.
3. Flux linkage ' λ '
4. Inductance $L = \lambda / I$

Solid cylindrical conductor

Assumption:

- Returning conductors are at infinite distance from incoming conductor so that there is no any flux linkage b/w the two.
- Current density inside the conductor is uniform i.e. there is no any skin effect.

$$\text{Flux linkage} = \lambda_{\text{inside}} + \lambda_{\text{outside}}.$$

$$\oint H \cdot dl = I_{\text{enclosed}}$$

$$H_x \cdot 2\pi x = I_x$$

$$H_x = \frac{I_x}{2\pi x} \text{ A/m.}$$



$H_x = \text{tangential to 'x'}$.

$$\therefore \pi x^2 \longrightarrow I A.$$

$$\pi x^2 \longrightarrow I \left(\frac{x}{r}\right)^2 \text{ for } x < r.$$

$$\therefore H_x = \frac{x I}{2\pi r^2} \text{ A/m.}$$

$$B_x = \frac{\mu_0 x I}{2\pi r^2} \text{ Wb/m}^2$$

$$d\phi = B_x \cdot A \quad \text{where } A = \text{Area} = 1 \text{ m}^2 = (dx \times 1 \text{ m}).$$

$$\therefore d\lambda = d\phi \frac{x^2}{r^2} dx$$

$d\lambda_{\text{inside}} = \frac{\mu_0 I}{2\pi r^4} x^3 dx \text{ Wb-turn/meter.}$

$$\lambda_{\text{internal}} = \int_0^r \frac{\mu_0 I}{2\pi r^4} r^3 dr = \frac{\mu_0 I}{8\pi} = \frac{1}{2} \times 10^{-7} I.$$

$$\lambda_{\text{internal}} = \frac{\mu_0 I}{8\pi} = \frac{1}{2} \times 10^{-7} I \quad \text{wb-t/m}$$

Now,

$$\lambda_{\text{outside}} = ?$$

$$H_x (2\pi r) = I$$

$$B_x = \frac{\mu_0 I}{2\pi r}$$

$$d\phi = B_x \cdot dx \cdot 1 \quad \text{Where } A = dx \times 1 \quad (\because \text{length of conductor} = 1\text{m})$$

$$d\lambda = d\phi = \frac{\mu_0 I}{2\pi r} \cdot dx = \frac{2 \times 10^{-7} I}{r} dx$$

λ_{12} = flux linkage b/w point D_1 & D_2 from the centre of the conductor

$$\lambda_{12} = \int_{D_1}^{D_2} d\lambda = 2 \times 10^{-7} I \ln \frac{D_2}{D_1}$$

$$\lambda_{\text{outside}} = 2 \times 10^{-7} I \ln \left(\frac{D_2}{D_1} \right)$$

$$\begin{aligned} \lambda_{\text{Total}} &= \frac{1}{2} \times 10^{-7} I + 2 \times 10^{-7} I \ln \frac{D_2}{D_1} \\ &= 2 \ln e^{1/4} \times 10^{-7} I + 2 \times 10^{-7} I \ln \frac{D_2}{D_1} \end{aligned}$$

$$\lambda_{\text{Total}} = 2 \times 10^{-7} I \ln \frac{D}{r'}$$

$$\text{Where } r' = r e^{1/4}$$

$$L_{\text{Total}} = 2 \times 10^{-7} \ln \frac{D}{r'} \quad \text{H/m}$$

$L \propto \frac{1}{r'}$, So, expanded ACSR reduces inductance bcoz r' is more in expanded ACSR.

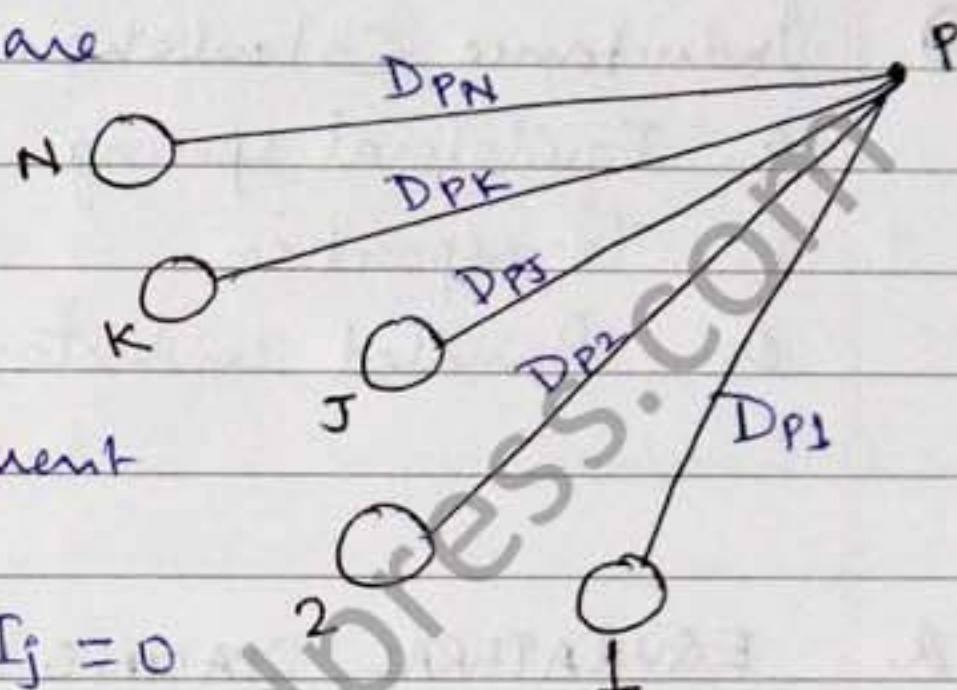
Composite Conductors

When large no. of conductors are taken in to an account

such that $\sum I = 0$ i.e.

In some conductor current is entering & in some conductor current is leaving.

$$I_1 + I_2 + \dots + I_N = 0 = \sum_{j=1}^N I_j = 0$$



Flux linkage ~~due to~~ conductor K at point P due to its own current.

$$\lambda_{PK} = 2 \times 10^{-7} I_K \ln \frac{D_{PK}}{r'_K}$$

Flux linking ~~due to~~ conductor K up to a point P due to current flowing in conductor K

$$\lambda_{PK} = 2 \times 10^{-7} I_K \ln \frac{D_{PK}}{r'_K}$$

Flux Linkage for conductor K up to a point P due to current flowing in J

$$\lambda_{KJ} = 2 \times 10^{-7} I_J \ln \frac{D_{PJ}}{D_{KJ}}$$

Total flux linkage = $\lambda_{KP} = \lambda_{KP1} + \lambda_{KP2} + \dots + \lambda_{KPN}$

$$= 2 \times 10^{-7} \sum_{j=1}^N I_j \ln \frac{D_{PJ}}{D_{Kj}}$$

Finally,

$$\lambda_K = 2 \times 10^{-7} \sum_{j=1}^N I_j \ln \frac{1}{D_{Kj}} \quad \text{wb t/m}$$